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IE Impulse

The magazine of the Austrian

Amateur aircraft builder

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Photo on the first page: Mont Blanc ©Masser

Imprint:
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Editorial

Dear aviation enthusiasts!

The 2020 flying season began so promisingly. I started my last editorial with that sentence. Unfortunately, these promises couldn't really be put into practice; the "corona crisis" had us in its grip for the entire season. But I don't want to complain; there were some beautiful moments this year too.
The Swiss EAS, like us, scheduled its annual summer meeting relatively late in August. This allowed us to take advantage of the time between lockdowns and enjoy wonderful experiences together with fellow aviation enthusiasts from Switzerland and abroad. Reports on these meetings can be found in this issue.

We're also receiving reports of significant construction progress; I just got a message from Hannes Maierhofer, one of the now rare Cherry builders, that his Cherry is ready for its stress test. This step is a major milestone for any home-built aircraft, roughly equivalent to passing a PCR test in the COVID-19 pandemic.

Furthermore, Christian Ellersdorfer sent us a report on the progress of his RV-10, which was even published in the Kleine Zeitung newspaper. Congratulations to the entire family for their exemplary cooperation on this unique project. After all, they want to take to the skies together!

In the vast majority of our home-built aircraft, the good old piston engine still serves as the power source. Many of these engines date back to the pre-war era; one might think that engine manufacturers have missed out on the developments that have taken place in vehicle manufacturing.
We know, of course, that the enormous costs of certifying a new aircraft engine are responsible for this, so existing engines are improved and upgraded, but the basic concept remains.

Now we have an innovative engine manufacturer in Austria who, drawing on their experience with agricultural and motorcycle engines and with the help of aviation enthusiasts, has dedicated themselves to the development of new aircraft engines. Initially, they successfully built two-stroke engines for the emerging ultralight (UL) movement, but soon ventured into the four-stroke class with the well-known Rotax 912. This was in the late 1980s, when my Cherry project had just progressed to the point where I needed to find a suitable engine. My then-building inspector, engineer Dundler, advised me against installing this new engine, as "we don't have enough experience with it yet." Since then, the 912 has proven itself exceptionally well, and its further developments have become classics used worldwide in the aviation community.

To further expand our experience with these engines and gain the latest insights, we invited Siegfried Heer, a specialist in aircraft engine development and testing at Rotax, to give a lecture afternoon at LOLH this year. Those who were unable to attend can read a short report about it here.

In keeping with this, Fritz Masser sent us an interesting travel report about his flight to Mont Blanc in his Cherry glider. He is, after all, the only one in Austria who has equipped his Cherry with the turbocharged Rotax 914 engine and can therefore boast remarkable flight performance.

Vortex generators and their effects have been known for years. Viktor Strausak, our old friend from Switzerland, reports on how critical flight characteristics of a high-performance aircraft were improved using these VGs.

With our aviation magazine as reading material, I wish us all a good and happy 2021! Beautiful, accident-free flights, very special flying experiences and... stay healthy!

Your chairman Othmar Wolf

Circumnavigating Mont Blanc twice...

...doable from Weiz in one morning


Above Trebesing, north of Spital an der Drau. View of Carinthia.

I've been waiting for two months for a high-pressure system from the Azores. I'd had Mont Blanc on my mind for a while. I wanted to tackle it in one day, flying there and back from Weiz. On November 6th, the day finally arrived. A high-pressure system from the Azores had been forecast for a few days. The weather, wind, and clouds were supposed to be okay. It would take at least 6 to 7 hours of flight time. In November, the days are considerably shorter. So, an early start. It's still dark on the way to the airfield. I arrive at 6:30 and first have to warm up my Cherry. The Rotax engine doesn't like the cold season very much, so it gets preheated for 30 minutes with an old hairdryer blowing on the cooling air intakes. Then, I fill up the tank. My flying buddy and friend Hans Haberhofer is already waiting at the fuel station this morning. "So, where are you headed today?" he asks me. I say, "To Mont Blanc and back." He: "And that can be done in one day?" - "Yes, God willing."

My Cherry has a 90-liter main tank and a 40-liter auxiliary tank located under the right seat. That makes a total of 130 liters. Of that, 125 liters are usable. According to my calculations, the flight should take approximately 6 hours and 30 minutes. With a 30-minute reserve, a flight time of 7 hours would be possible. In a-

The Rotax 914 consumes about 17 liters per hour in cruise flight and reaches a cruising speed of 130 knots. As a precaution, I've listed Lienz as an alternate airport in the flight plan – in case we run low on fuel.

I've always wanted to try and see how long my glider can stay airborne and how far I can get on 130 liters of fuel. I submitted the flight plan to Austrocontrol the day before.

I also had to take oxygen with me, because I wanted to fly over Mont Blanc, which, at almost 5000 meters, is known to be the highest mountain in Europe. With my Turbo Cherry, climbing to 16,200 feet shouldn't be a problem.

Departure at 8:06 local time. The flight was planned from Weiz via Graz to Lienz, Brixen, Bolzano, Samedan, Maloja Pass, Domodossola, Lugano in southern Switzerland, Zermatt, Matterhorn, Chamonix, Mont Blanc, and back to Weiz. The weather is fantastic. I set the altimeter pressure to 1035 millibars; it can't go any higher anyway.

Not a single cloud in sight. However, at this time of year, one has to expect high fog in most valleys in the morning. And that's exactly what happens.

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In Carinthia, all you can see are the mountains, and from South Tyrol onwards, the visibility improves. In southern Switzerland, to my right, everything is clear, and to my left, the Po Valley with Milan shrouded in high fog up to about 2,500 feet. A sea of clouds as far as the eye can see. With a few minor detours along the way (danger and restricted areas), I can fly directly to almost all the points on my flight plan. I should also mention that I like to fly my plane at a higher altitude. Mostly between 8,000 and 10,000 feet. Not because I don't trust the Rotax engine, but because it's a passion of mine to view the world from a bit higher up. I've already passed the Matterhorn at 13,000 feet and still have to look up to see the summit.

The Matterhorn from below

The view down into the valley and towards Zermatt is fantastic. Mont Blanc is still 70 km away. I could already see it, but I'm still much too low. So, full throttle and...

We climb to 16,500 ft. While radioing and looking around, I overlook the fact that my plane is still climbing, and before I know it, I'm at 17,000 ft. The Cherry would climb even further, but that's not necessary. I'm flying


Maloja Pass with views of Samedan and the airport

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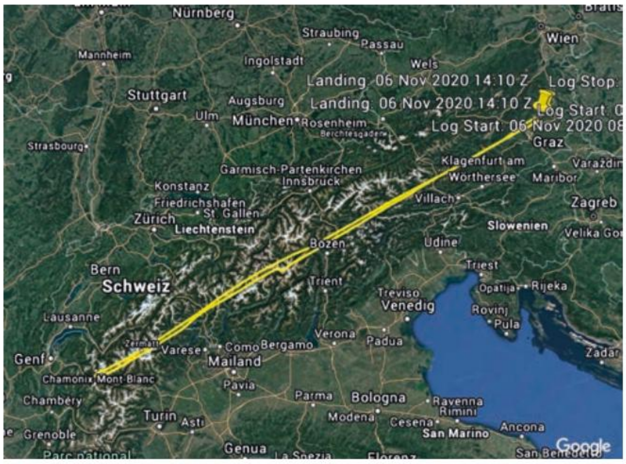
Above Chamonix - view of Mont Blanc

We're already above the summit of Mont Blanc. The view up here is indescribably beautiful, like a postcard. After two circuits around the mountain, we head back to Weiz. Landing in Weiz (LOGW) at 2:10 PM. I need a few days to mentally process it all. It was simply an amazing experience.



The flight details:

Aircraft: Cherry BX 2
License plate: OE-CMF
Engine: Rotax 914 turbo
Flight time: 6 hours and 4 minutes; **Total flight distance:** 1,399 km; **Maximum altitude:** 16,911 ft
Top speed: 275 km/h
Average speed: 230 km/h
Consumption: 103 liters of mogas (16.8 liters/hour)

PS: I'm already planning my next flight. Next time I might not turn back, but fly on. Mallorca should easily be doable in one day. I still need to figure something out about the toilet situation.

Fritz Masser

Photos: ©Masser

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Vortex Generators in Practical Testing

Since the Lancair Legacy with a 210 hp engine was found to be 3 decibels too loud for the lowest noise class A in the noise measurement in Switzerland, but the same aircraft with 310 hp achieved noise class D, I am now trying to reduce the noise in a completely unconventional way with vortex generators and extensive test flights.

I should mention that I've never flown a Lancair before. Based on what Lancair pilots have told me, I expected extremely difficult and demanding flight characteristics. Before my stall test flights, Legacy pilots warned me about the very dangerous stall behavior, especially at full throttle.

I was surprised by the high approach speeds of initially 90 kt and not below 80 kt in the final approach (according to AFM), as otherwise the roll angle can no longer be controlled.

I approached the test flights with the beautiful Lancair Legacy with great respect and caution.
As with all my stall test flights, the first thing I did was glue wool threads to the wing. This allows me to observe the airflow behavior over the wing throughout the entire speed range up to a full stall, initially in its original state, without vortex generators.

First test flight without VGs

It became clear to me during the first stall why Lancair pilots force their aircraft onto the runway at 80 knots without attempting to fly out. The ailerons are poorly designed and receive hardly any airflow below 80 knots, thus losing their effectiveness and function, which is very problematic in gusts.

It's dangerous. Even at 78 knots, I made full aileron deflections without the wing moving – the ailerons lost their effectiveness and therefore their purpose.

During takeoff, the aircraft can only be controlled with the ailerons from 80 knots upwards. Fortunately, this dangerous phase of flight is very short.

I flew in all configurations, from idle power to maximum power, up to a full stall, including gear down and full flaps. I was advised against stalling at maximum power, as it could become unpredictable. I did it anyway and survived!

Based on my initial assessments and experiences with the beautiful and elegant wing shape, which unfortunately is not aerodynamically optimal in my view, I expected this dangerous stall behavior even before the first flight. My suspicions were confirmed.

For attaching the VGs, we fabricated a special jig for this wing shape, based on my own calculations, with smaller than usual spacing of only 50mm and the angles precisely aligned with the flight direction. With this, we initially glued only the outer...

ren VGs are positioned far forward at 7% of the wing chord in front of the aileron.

Additionally, we mounted VGs under the elevator with a 30 mm gap and 100 mm in front of the hinge.

As with many aircraft, the elevator control is too small and has too little effect when landing at low speeds.

The second test flight

Using VGs only on the outer wing half, in front of the ailerons, was a complete success. The aircraft remained fully controllable until a full stall, and the stall speeds were significantly lower in all configurations.

We then glued the VGs (airflow control elements) to the inner wing half, spaced 90mm apart. The larger spacing of the VGs ensures that the airflow separates on the inner wing before the ailerons become ineffective.

Third test flight

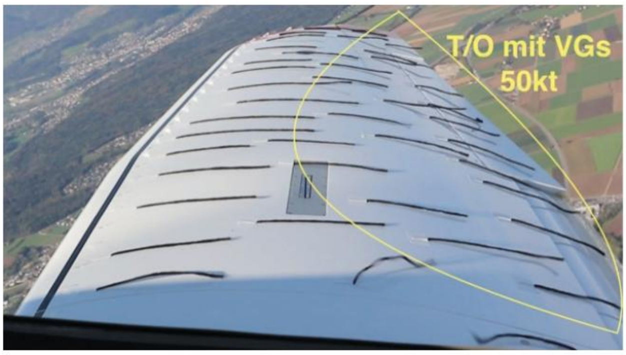
I was impressed and thrilled by the success; my expectations were far exceeded!

The aircraft is always docile in the stall, control-

The bar and ailerons remain effective at all times. We were able to reduce the stall speed to 15 knots (-22%) while maintaining full aileron control!

The touchdown speed was 80 knots without VGs, now it's 55 to 60 knots. A reduction of 20 to 25 knots (-30%)!!!!

The landing distance has thus been massively reduced to under 400 m.



Changes with VGs:

Rotate before takeoff even before reaching 60 knots and initially climb at 65 knots. New takeoff distance approximately 300 meters.

Without VGs, the aircraft was uncontrollable below 80 knots.

The perceived rate of climb increased by approximately 200 ft/min, while the cruise speed remained unchanged.

These values still need to be verified.

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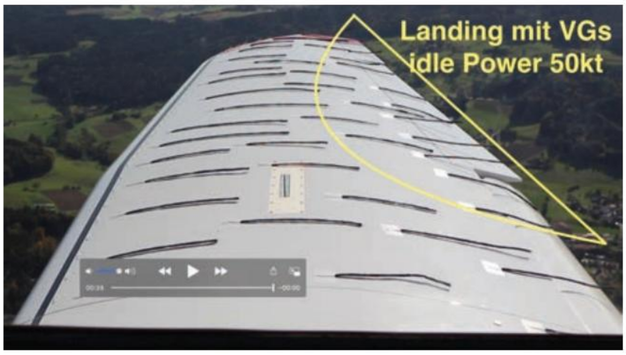
Landing configuration: perfectly controllable at 50 knots! (Previously 80 kt)



For me, the Lancair modified with VGs has become a very docile and easy-to-fly aircraft that can be operated without problems from relatively short runways. Thus, the Lancair has become a "normal" aircraft in terms of handling, which is significantly easier to fly than, for example, a Kitfox with a tailwheel (takeoff and landing).

A great start at 44 knots, perfectly controllable! (Previously 80 kt)

I hope that as many Lancair pilots as possible will use our experiences to turn their Lancair Legacy into "proper" and safe aircraft.

For such expensive aircraft, the effort of approximately one day's work and material costs of $162 should not be an issue.

Position of the VGs:

Leading edge of the VGs at 7% of the wing chord. Distance between inner wings 90 mm.

Outer wing in front of the aileron: 50 mm clearance.

The angle is always 15 degrees, both left and right when viewed in the direction of flight. The outer wing, which has a trapezoidal shape, must be corrected by 7 degrees.

Under the elevator, 100 mm in front of the hinge, distance 30 mm.

Changes:

Approach speed 65 kt, -25 kt (-27%) Landing speed 60 kt, -20 kt (-25%).

Takeoff roll now approximately 300 m, landing much shorter than before, under 400 m. Stall down to 15 kt (-22%) lower.

The aircraft remains controllable and very docile even in a stall.

My recommendations on how to fly a Lancair with VGs: Every pilot must test and determine these values themselves with their own aircraft!

Start:

Controlled and fine rotation at 58 knots, then an initial climb at 65 knots. For takeoff, trim approximately 3 mm nose down for the light 210 hp engine. Takeoff roll approximately 300 m!

Landing:

Downwind clean at 90 knots. Basic gear and flaps down speed initially at 75 knots with a bank angle. Stabilized final speed 65 knots, with short final practice even 60 knots possible. Caution: Reduce power late and rotate immediately, as speed drops rapidly. Risk of short landing and hard landing!

Result:

Landing distance under 400 m!

We had a blast during the test flights, and the results far exceeded my expectations. The test flight program now needs to be repeated so that the noise test can be conducted again using the new data.

The vortex generators are from STOLSPEED Austria. www.stolspeed.com It takes approximately 200 VGs for $162.

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Photos: ©Strausak

Why can a layperson build an airplane... ...but not a car?

The question is certainly valid. You won't find the answer on the internet. It's not entirely straightforward, however, as it involves factors such as psychology, regulations, safety, physics, and economics. To start with the last one: both cars and airplanes are typically manufactured in factories, but in fundamentally different ways.

Robots dominate car production, welding better and more precisely than humans. And above all, cheaper. A car has become the classic mass-produced item, with a galvanized body, but designed to be discarded after seven years. (Seven years? That's how long it takes for the average person to go from buying the first parts to getting the car registered.) Mass production makes the car not only a mass-produced item but also a cheap one. DIY car builders could never compete on price.

Airplanes, on the other hand, are expensive; however, self-built ones are usually less expensive than industrially manufactured ones. In the case of aircraft production, "industrial" means primarily manual labor. There simply isn't enough demand for millions of units, and therefore no demand for industrial robots. Besides, seemingly primitive tasks like driving rivets aren't easily automated because, in many cases, two workers are needed for a single rivet: one with the hammer and one who (often inside a fuselage or wing) holds a block against the rivet being driven.

The robots aren't that advanced yet. Ultimately, it probably all comes down to a question of liability.
Workers can be encouraged, motivated, and even monitored to ensure they don't make certain mistakes. Robots, as we know, are perfect. If they make a serious error, then all the pieces in a production run will quickly have it. The recalls in the automotive industry are quite revealing in this regard.

Strictly speaking, it's all about security.
In this respect, road traffic differs from aviation. The automotive lobby and relevant authorities define safety as the highest possible probability of survival for the occupants in an accident. This is addressed with each model individually.

Tested by realistically crash-driving lifelike male dummies into an obstacle in a production vehicle. This is how, after several decades of steadily increasing production, we arrived at today's ubiquitous SUV, which even in rampage situations still offers its occupants sufficient "safety." Superficially, a car also appears safer as its weight increases: it can only be crushed by a truck. In reality, the danger increases with both mass and speed. And the "safety" of the car offers no help whatsoever to the cyclist or other person hit by the speeding vehicle. The automotive industry is therefore far less concerned with preventing accidents than with increasing the occupants' chances of survival. Airbag sales are a business, and so is mitigating the consequences of accidents.

Airplanes, on the other hand, have no airbags or armor plating. Anyone who crashes a four-seater into the concrete wall of a hangar at 50 km/h is unlikely to survive. No airplane is equipped to withstand such accidents, nor do they need to be. In this sector of aviation, the probability of an accident being fatal is so high that it's wiser to do everything possible to prevent it rather than mitigate its consequences. Ultimately, any measure to cushion the impact would make the aircraft heavier. And in aviation, weight is inherently conducive to accidents. That's why each of us has the weight and balance diagram of our aircraft memorized.

Cars that go out of control and leave the road don't necessarily end up against some rigid obstacle, but often simply in the grass. That wouldn't necessarily be a cause for injury or even death.

Aircraft in an uncontrolled state are no longer controllable. While they may "land" in a field, if this happens at low altitude, they will hit it at a right angle and with increasing speed until the very end. This is precisely the situation that must be avoided at all costs.

First, it is ensured that the main tholm can withstand even the most severe stresses of a flight.

If the aircraft can withstand the stress (our machines usually have to prove this with a load test), then the engine mount and landing gear are subjected to the same procedure. This ensures virtually all the safety features an aircraft can offer. Now, only the pilot can cause an uncontrolled flight condition. But that's no different from the safest car. If someone misses the opening in a highway tunnel, all the built-in safety features won't help them.

This naturally has implications for building your own car or airplane. A car builder would have to incorporate crumple zones, collapsible steering columns, yielding instrument panels or glove compartments, airbags, and side-impact doors. These requirements far exceed the capabilities of most DIY enthusiasts.

The builder of an aircraft, on the other hand, must first and foremost adhere precisely to the designer's instructions, and then carry out the load tests (unless a colleague has already subjected an identical aircraft to these tests). That's all there is to it. The design itself has already been approved or rejected by the authorities before construction even begins.

In any case, our comparison is flawed in one crucial aspect. We are not comparing the self-building of an airplane and a car per se, but rather the self-building of an airplane and a fast car. Every car (unless it's a disguised moped) can drive on the highway, and there it must be able to travel faster than 40 km/h, fully loaded and uphill.

And if it's capable of that, it can easily reach 100 km/h on a flat country road. But it also has to be able to withstand that speed, even if the road suddenly has potholes.

Potholes in the road, however, pose a completely different kind of stress than "air pockets" (vertical wind shear, turbulence). The wheel suspension and chassis must be designed to handle this. At 100 km/h, the design and setup of the suspension also play a crucial role if the car is to remain spun out of corners prematurely. This is practically impossible for a layperson to achieve. In contrast, assembling a wing according to a template (without shock absorbers, as these wouldn't help against turbulence) is not rocket science.

Airplanes, on the other hand, only encounter potholes on grass runways (where they are less severe).

How it rolls (than in asphalt) and how it corners on the ground is irrelevant. Essentially, it must be able to roll straight for the few hundred meters of takeoff and landing distance. During takeoff, it's pulled by the propeller, similar to a toy duck being pulled by a small child. Most importantly, none of the wheels are driven. Achieving this isn't a difficult feat for the designer and builder. An airplane doesn't typically accumulate thousands of kilometers of mileage over its lifetime.

It's rarely – at least in our case – a means of transport, but primarily a machine with which one fulfills a dream: to fly. It's meant to be enjoyable, and usually it is, as evidenced by the fact that there's significantly less swearing at the control sticks than behind steering wheels. In an airplane, one enjoys beautiful days, but not rain and hail. Therefore, airplanes don't have to meet the same high standards in this regard as cars. Cars are meant to protect their occupants from all weather conditions; airplanes only need to do so in exceptional circumstances. There's practically no weather that can prevent one from driving a car. However, pilots are familiar with many meteorological situations that they prefer to observe from the ground.

Significant differences also exist in comfort for the driver or pilot. The average car is comfortable, well-sprung, and quiet. Airplanes, by their very design, cannot offer any of that. Therefore, aircraft manufacturers can save themselves the considerable expense of a soft landing gear or effective noise insulation. Ideally, one flies with a headset anyway.

A crucial argument is the weight difference between cars and airplanes. It makes a big difference whether you're dealing with heavy steel components or aluminum or carbon fiber structures. While airplanes also need an engine, they don't require a transmission, a differential, dual-circuit braking systems, power brakes, or exhaust aftertreatment. This saves a considerable amount of weight. Furthermore, the materials used in aircraft construction are much easier to work with than sheet steel.

For all these reasons, the possibility of building a car yourself is limited to buying a chassis and manufacturing a suitable body.

However, usable chassis are no longer available because cars today are only built with self-contained suspension.

Load-bearing car bodies are produced. So you're dependent on buying an old chassis from a VW Beetle or Citroen 2CV. That doesn't sound very exciting. Besides, with the resources of a DIY store consumer, you can hardly put together a body like one from Bertone or Pininfarina. After all, you also need the right windows, and those have to be made of safety glass. Aircraft manufacturers can (and must) use the easily workable acrylic glass.

men.

There are some very elegant aircraft designs, such as the Mooney or Falco. Surprisingly, similarly elegant machines can also be found in the homebuilder scene (Lancair, Pulsar, Cherry). These shapes can easily be achieved using aluminum, fiberglass-reinforced plastic, and wood. Extravagant features like tail fins, spoilers, and wide tires would be neither practical nor permitted on an aircraft.

In any case, the majority of home-built aircraft are superior to industrially manufactured models in at least one or two disciplines.

This isn't surprising, as industrial competition is quite limited in terms of model variety. We're essentially only competing with two-seaters (four-seaters can be built in-house, but the regulatory requirements for such machines are so stringent that most people don't want to put themselves and their wallets through that). And these competitors are generally significantly more expensive than our machines and, in many cases, slower as well.

We are allowed to repair our own machines. Off-the-shelf models must be taken to a shipyard for every single modification. We can install all sorts of parts (apart from safety-related exceptions), whereas industrial aircraft must be equipped exclusively with aviation-approved parts.

And those things cost a pretty penny. An aircraft is considered self-built if the er-

The builder must have performed at least 50 percent of the necessary work himself. Pure assembly is therefore not permitted. With a car, this would often be impossible, because a purchased VW chassis is, strictly speaking, already a functioning car – just without a body.

Aviation is probably the most conservative of all branches of technology. While it does accept new, improved solutions, this only happens after numerous and, above all, extremely expensive safety tests.

(The Boeing 737 MAX is a notorious exception.) Any changes for reasons of design, fashion, or marketing are not approved by the authorities for good reason.

A standard size for aircraft instruments

Therefore, the component manufacturers haven't even attempted it. An engine that has proven itself over several years is only modified if absolutely necessary. The instruments are standardized, and have been since the war. If I want to install one with an 80 mm diameter, I don't need to worry about receiving one with a 78 or 85 mm diameter. I could probably install several instruments from a decommissioned Concorde in my aircraft. Every radio comes with absolutely foolproof installation and operating instructions.

The first pioneers who wanted to build an airplane in Austria had the same experience as all those who demand something unprecedented from a government agency. But with persistence and the good fortune that some young officials were more curious than opposed to the idea, our association succeeded.

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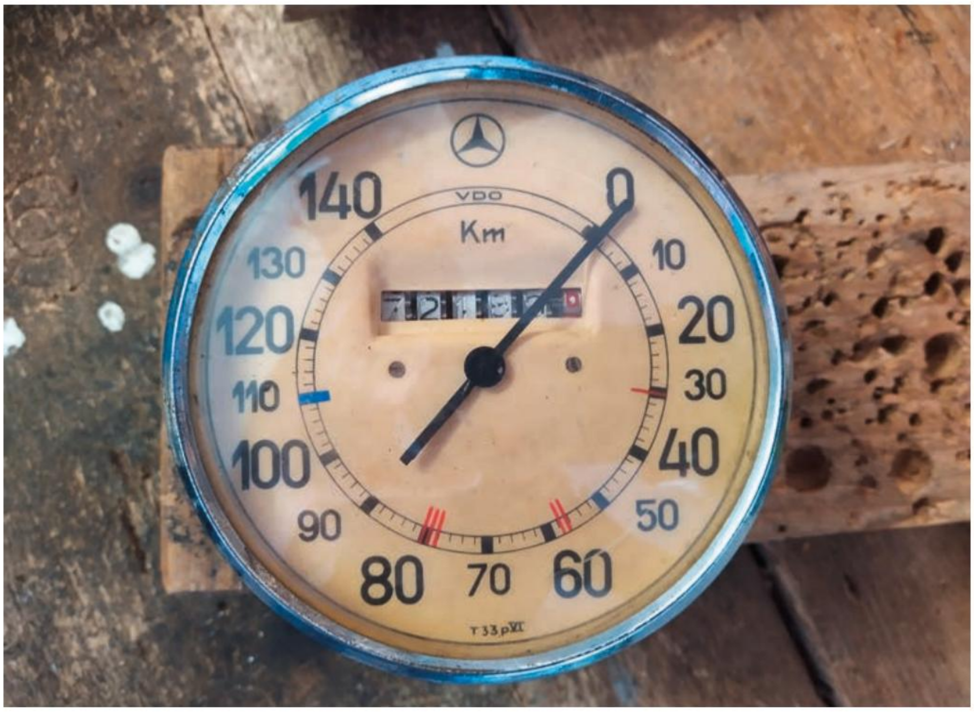

Cars once had easily replaceable speedometers. Photos: ©Baumann

If you're thinking about building a car, then try simply buying a speedometer. Not one for go-karts or motorcycles, but one for cars. They do exist for airplanes – increasingly from China – but you can buy them and they'll fit.

Ultimately, you'll soon find yourself stuck in traffic with your self-built car. Simply lost in the crowd.

That hardly ever happens to us. On the contrary: every now and then a controller asks for the type of aircraft I have, and because he doesn't know it, he has me describe it. The sky's too vast for a traffic jam, and there's only a mass of 50 small planes on the ground when the Igo Etrich Club holds its annual meeting. Then, others regularly steal the show.

Founder Rudi Holzmann was the first to fly his KR 2. The authority, in our case Austro Control, has developed procedures over the years that allow even a layperson to realize their project. The necessary inspections are considered rigorous by foreign standards.

For this reason, experimental aircraft, which are approved with the red-white-red flag on their tail, are just as safe as factory-built aircraft.

Christoph Canaval

What the Rotax engine loves - and what it doesn't

You can't build an airplane without eventually stumbling across the Rotax 912. It's simply THE engine for two-seaters, whether in the standard 80 hp version (which is the least temperamental), the S variant with 100 hp, the 914 with a turbocharger and 115 hp, or the 915 IS with a supercharger and fuel injection and 141 hp. These engines are relatively modern, lightweight, and have been on the market for quite some time – meaning a considerable amount of experience has already been accumulated with them. These experiences are a regular topic of conversation among our fellow enthusiasts.

In September, engineer Siegfried Heer, who played a key role in the development and testing of the engine series, gave a lecture in Hofkirchen, from which we (for all those who were not there) record the most important tips.

All engines are available in certified or ultralight versions. Technically, there is no difference between the two; the former simply have the necessary documentation for certified aircraft.

Anyone considering buying a 915 IS should look for a robust propeller. The commonly used models from Woodcomp or Neuform are no longer suitable for this engine; "they lose their blades. This engine is not a toy."

Fuel delivery in the lines is impaired if vapor bubbles form (or can form) there. This problem is becoming increasingly significant because fuel companies are supplying gasoline with the highest possible vapor pressure, tailored to modern vehicle engines. However, this poses a significant challenge in aircraft.

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increasing problems with vapor bubbles that form at high outside temperatures and high altitudes.

The aviation fuel (avgas) prescribed for almost all other aircraft engines therefore has an extremely low vapor pressure. (In winter temperatures, one could hold a lit match over a bowl of avgas and the fuel would not ignite.)

Rotax recommends "Shell Sport Plus 4," a semi-synthetic oil. Mineral oil is required for the lead additive. Using lead in a car would damage the exhaust system. The oil can easily remain in the engine for 200 hours, or even 300 hours without avgas operation. However, with pure avgas operation, the oil must be changed after 25 hours. Liqui Moly advertises with the slogan "Become a test pilot!"

The use of avgas is expressly not recommended for Rotax engines. However, if it is unavoidable for specific reasons, the oil must be changed more frequently.

To avoid problems with vapor lock from the outset, careful installation of the return line is essential. For this reason, fuel pumps in fuel-injected engines are also significantly oversized. If the required fuel quantity is a maximum of 20 liters per hour, a pump with a flow rate of 130 l/h is used.

Standard built-in.

All Rotax boxer engines with carburetors always have two. The manufacturer designed it this way to prevent total engine failure. With one carburetor functioning, the 912 S still manages a respectable 34 hp. It's certainly enough to get the aircraft to the nearest runway.

The exhaust springs (which keep breaking) must be secured with wire (preferably with two wraps). This simple method prevents vibrations that regularly cause the springs to break. The heat from the exhaust doesn't affect them.

Occasionally, you hear suggestions like: Fill the springs with heat-resistant silicone. This is about as effective as filling them with concrete. Springs "secured" in this way have already led to engine fires.

The coolant should contain Glysantin. Anything else will destroy the shaft seals. The engines run for 45 minutes without coolant (at low power, of course).

However, this oil is not suitable for Rotax engines. An aircraft engine must not be able to be overfilled. That's why Rotax engines have external reservoirs for both coolant and oil.

The engines must be able to withstand a fire for five minutes – this is also tested in practice. A crucial aspect of the tests is that no fluid leaks out.

At the end of the presentation, a video was shown. It depicted a running Rotax fuel-injected engine, whose connections to the various sensors were disconnected one by one until only a few remained intact. The engine continued to run nonetheless. This illustrates the redundancy of engine technology, which, while requiring considerable effort, clearly also provides a high degree of safety.

Christoph Canaval/Othmar Wolf

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Igo Etrich Meeting 2020

Arado 96 by Alois Krennmeir

Preparations for our annual Igo Etrich meeting in August begin long before the actual event date. This includes, in particular, the question of the optimal venue.

After being unable to come to Krems for several years for various reasons, my inquiry this year was immediately met with a positive response, and we were warmly welcomed back. The decision was made: Krems LOAG would be the venue for 2020. Only the usual details needed to be finalized, such as parking areas and landing fees.

ren etc.

The date of our meeting has followed the same pattern for years: the second full weekend in August. We've had good experiences with the weather at this time of year, which is a basic requirement for a successful event. The second advantage of this fixed date is that everyone interested, both at home and abroad, can prepare for it well in advance and plan accordingly.

Unfortunately, to make matters worse this year, dark clouds in the form of the Corona pandemic appeared as early as spring.

The Cri Cri by Hans Fischer

We were still able to hold our annual general meeting as usual at the beginning of March, but then things moved very quickly with the cancellation of all flying events: Aero Friedrichshafen, the meetings in Germany, Sweden, England, all cancelled due to Corona restrictions.

gene.
However, by summer restrictions had been eased again, so we continued our planning.

The Ollfisch Luciole

Guided tour. Airshows are generally open-air events, so there shouldn't be any problems. The only potential issue is the catering in the guest area.

Stan could be served in accordance with the regulations.

The weather forecast for our weekend was promising, so the first participants arrived on Friday morning. This allowed us to set up our headquarters using a special trailer provided by Sigi Schicklgruber. As always, I was able to enlist Heidi as our head receptionist. However, she didn't have an easy job this time, as it was swelteringly hot during the day. The airfield owner had provided ample shade for his outdoor areas, so most people were able to seek refuge there.

This time, we have been assigned an area quite far to the east as a parking space, so as not to disrupt regular flight operations.

Hellmuth Lehners UL Sunwheel

Camping next to one's own aircraft was also permitted, so everyone could be satisfied.

By Friday evening, 31 aircraft were had arrived, including some new faces alongside all the familiar faces from neighboring countries. For example, the eccentric "Olle" Ollfisch with his F-registered Luciole from Germany. Unfortunately, one of ours arrived...

The economy was a factor of uncertainty, but the manager of the Fly restaurant promised to come up with something so that our guests would have enough food.

Dutch Cherry (Andre Gremmen)

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RV 9 A by Lasse Espersen

A visitor from Germany was injured in a botched landing with his HB207: the nose landing gear collapsed and the propeller was damaged. The pilot himself was unharmed and was able to travel home the same day using an affordable ride to arrange his return transport.

We hadn't expected such a large turnout for the reasons mentioned above, but the trip to the wine tavern in Langenlois for dinner was also excellently organized by several volunteers, including some pilots' wives, and everyone found a seat in the end. The traditional wine tavern snack was delicious, and as always, there was plenty of aviation talk. My seatmates, a Danish couple, turned out to be the builders of an RV-9A, who now live in a flying village near the Channel coast in France. Lasse Espersen told us their story and even invited us to visit them at their flying club, which we gratefully accepted. Finally, we had to prepare for the return journey to our various accommodations.

Saturday is the day when most visitors arrive, and once again, aircraft came from Germany, Switzerland, France, the Netherlands, Slovenia, and of course, many from Austria itself. The Swiss, with their EAS President Werner Maag, had set up their own HB area on the parking apron. Vice President Detlev Claren visited us as a representative of the German OUV, though this time not in his familiar Citroën 2CV LI-1, but by car due to health reasons. Thank you, Detlev! Among our members, Alois Krennmeir deserves special mention again; despite his advanced age, he still arrived in his beautiful Arado 96. Model turbine specialist Peter Jaka-

One of the longest-serving home-built aircraft builders: Hubert Keplinger

Thankfully it was only minor damage.

It is likely that some inaugural flights had to be postponed this year due to the coronavirus.

On Sunday morning, a first glance out the window or from the tent revealed good flying weather for the journey home. Most treated themselves to a leisurely breakfast in the airfield restaurant or a pleasant chat with fellow pilots before preparing their gliders for the flight home.

dofsky brought his Alouette III, the real thing, not a model. Andre Gremmen, well-known in Cherry circles, came from Holland with his PH-TBJ, and new to the event was round-the-world pilot Hellmuth Lehner, this time with his new Sunwheel ultralight. Hans Fischer also brought his eye-catching Cricri OE-FFW again and demonstrated with two flights that it is now truly ready for flight.

By evening, we had counted 75 aircraft – an incredible success despite the obstacles. We met for dinner at the airfield restaurant, adhering to hygiene guidelines, hoping everyone would find a seat. The owner's promise was well-founded; most were able to sit outside, and everyone enjoyed an excellent meal. A big thank you to the team at Restaurant Fly! Unfortunately, prizes for completed aircraft could not be awarded this time.

All in all, another successful Igo Etrich meeting in 2020. A big thank you to the team and the managers of Krems Gnixendorf airfield, the owner of the Fly restaurant and his team for the excellent hospitality, and to all arriving and departing pilots for their discipline.

Thanks also to my Heidi, who once again organized everything perfectly and who is always there not only for our family, but also for the Igo Etrich family.

Othmar Wolf

Photos: ©Gerhard König

Shark by Walter Prossinger

Impressions from Switzerland

The EAS meeting was one of the few that took place in 2020.

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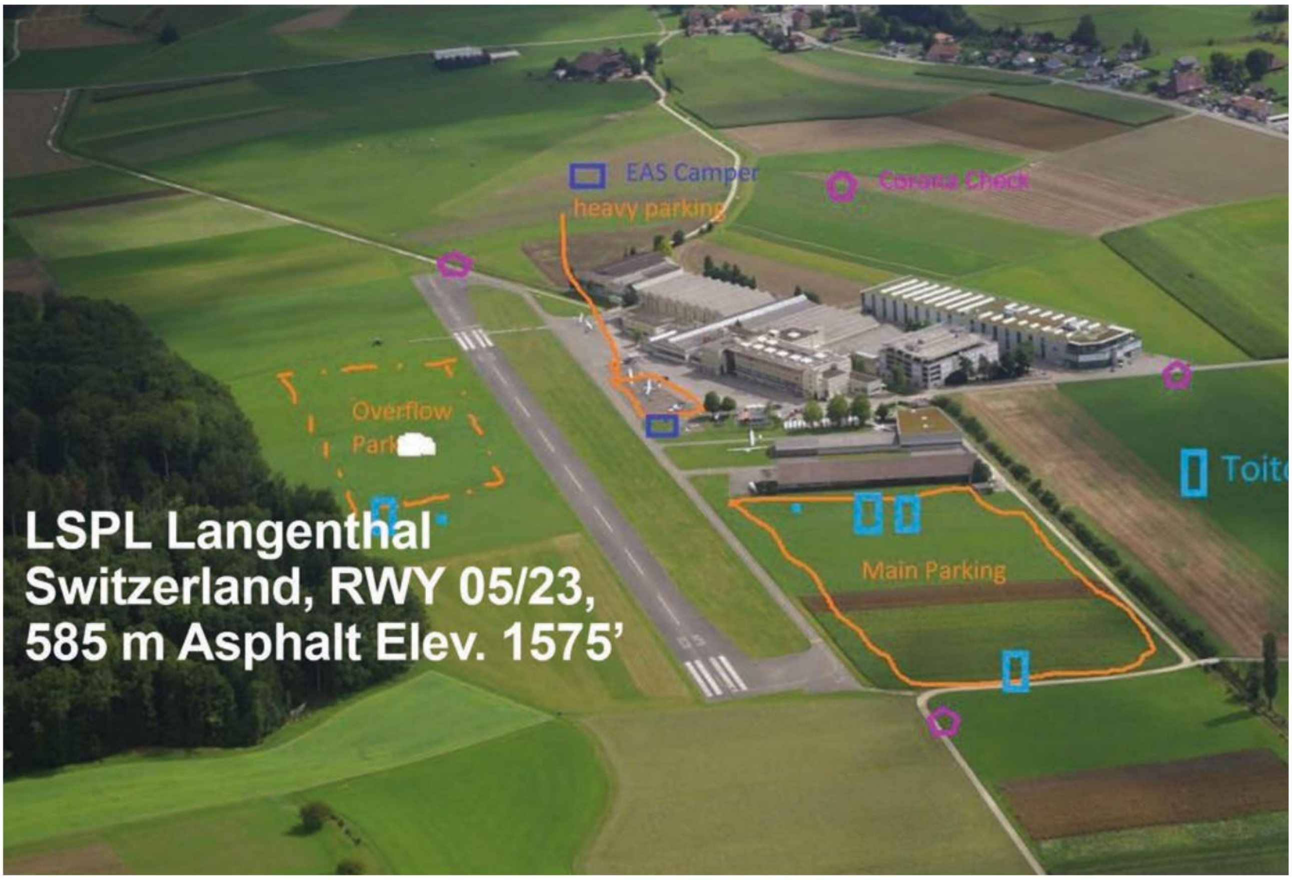

LSPL Langenthal
Switzerland, RWY 05/23,
585 m Asphalt Elev. 1575'

10


101-LI

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Christmas, as it used to be...

This year (I'm writing this in Advent 2020) everything is different! The calendar may indicate a quiet Advent season, but this year, the pre-Christmas joy just isn't there. The year 2020 has been largely characterized by worries, both health-related and economic, and above all, by fears.

While the first lockdown was accepted with some resignation and as a matter of course in the fight against COVID-19, people now have much more fear, many more horrors, and above all, much more despair and discouragement.

The propaganda machine surrounding illness and death is working at full capacity. Hasn't all this been done before?

On the other hand, resistance is stirring right now, resistance against all these regulations, whether one considers them sensible or not. That's a matter for each individual to decide. In any case, it's interesting that even the Supreme Court is dealing with these infringements on our fundamental rights…

Common sense is needed again. Many are starting to wonder why such a disease is occurring now. Didn't we read somewhere about the seven plagues? Or why did Atlantis disappear? Pride comes before a fall.

Have we paid too little attention to the signs of the times? Have we lived from day to day? Taken everything for granted?
Have we taken it? Have we completely forgotten about our immense consumerism?
Have we perceived the subtle, interpersonal vibrations and needs around us?

You might see Corona as a worse flu, or as a deadly plague, but that doesn't really change the "cause" or the indication. As long as we continue to live as we do...

Living now, there will always be such warnings, whether they're called Corona or something else. We have a choice.

The time is ripe for change, change for the better. We can either participate, help shape it, and live accordingly, or there will simply be no place for us on this planet. That's how evolution has worked for thousands of years.

If we are aware of this, truly aware, then we will once again experience "Christmas like in the old days." If we are capable of truly embracing the meaning of "Christmas."

(These are just my personal thoughts, without meaning to offend anyone.)

Heidi Wolf

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Lavanttal, Thursday, September 17, 2020

A dream, who entered the world leads out

By Martina Schmerlaib

Christian Ellersdorfer (35) from St. Andrä dared With the help of his partner and his father, he accomplished something extraordinary: the construction of an RV-10.

There's quite a lot you can do. It undertakes to create long-cherished dream to realize. In the case of the two Lavant Valley residents, Christian Ellersdorfer (35) and Lisa Schrofler (28), even a great deal. The ambitious plan: to build a four-seater, airworthy propeller plane – an RV-10. They had to do more than just that. a dust-free and heated build a workshop, but also to obtain a building permit apply to Austro Control and high transport costs to take on the burden of coming from America. But their dream is such a Much closer.

"We are already working on the plane." for about two and a half years. Each of us has three so far. 1200 working hours in the "spent time in the workshop," says the St. Andräer, who flew the plane together with his partner Lisa and her father Gerhard in the builds his own barn. The basic structure of the aircraft is now complete; now it's time to work on the Interior fitting including avionics as well as the installation of the engine. But until construction could begin, there was a Extensive research is required. "I am featured in a professional journal I came across an article by an aircraft manufacturer. Since then, the idea has fascinated me. could not be let go, even wanting to build an airplane. I then searched the internet and inquired in all media and researched extensively," says Ellersdorfer, who Finally, all the individual parts for the plane from the Americas-niche company "Van's Aircraft" received. And those were So many that the manufacturer hasn't even counted them yet. "In the horizontal stabilizer alone, we have over 1000 "Rivets were used," says Schrofler. First, the individual pieces were digitally recorded and cataloged. The process was meticulous. Parts were sorted, counted, numbered, and stored on shelves and in bins. The assembly instructions arrived in five separate folders. Included. "You can imagine it similar to an Ikea assembly kit "Imagine, only much more complex."

says the assistant professor at Institute for Vehicle Safety at TU Graz, which is dedicated to Dream of owning your own airplane He financed it himself with his partner. He doesn't reveal how much it costs, but "you can afford it". imagine that a simple Wooden planes cost about as much as a car costs. Upwards There is no limit."

After the building permit with The three have many editions Idealists got started and took a step Each part was installed step by step. Accuracy and expertise

Left: Gerhard and Christian Ellersdorfer. Right: After the shipping the parts by Delivered by truck KK/PRIVATE(5)

Gerhard and Christian Ellersdorfer.

WHAT, WHEN, WHERE?

The most important dates from your region

TODAY, 17. 9.

WOLFSBERG. "The Gruffalo." Puppet theater of the world-famous children's book character The Gruffalo. Performed by the Hatschi Puppet Theatre. Kleinedling Market Grounds, 16 o'clock. Tel. 0660-187 31 76

WOLFSBERG. Alcohol counseling and aftercare group. Professional, Confidential individual and group discussions. Every other Thursday. AVS Social and Health Center, 7 to 9 pm o'clock. Tel. (04352) 51 51 20

PHARMACY

WOLFSBERG. team santé a[illegible] pharmacy, Klagenfurter Straße 35. Tel. (04352) [illegible]

We are here for you in the Lavant Valley

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Fri. 8 a.m. - 1 p.m.

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Subscription service: (0463) 5800 100

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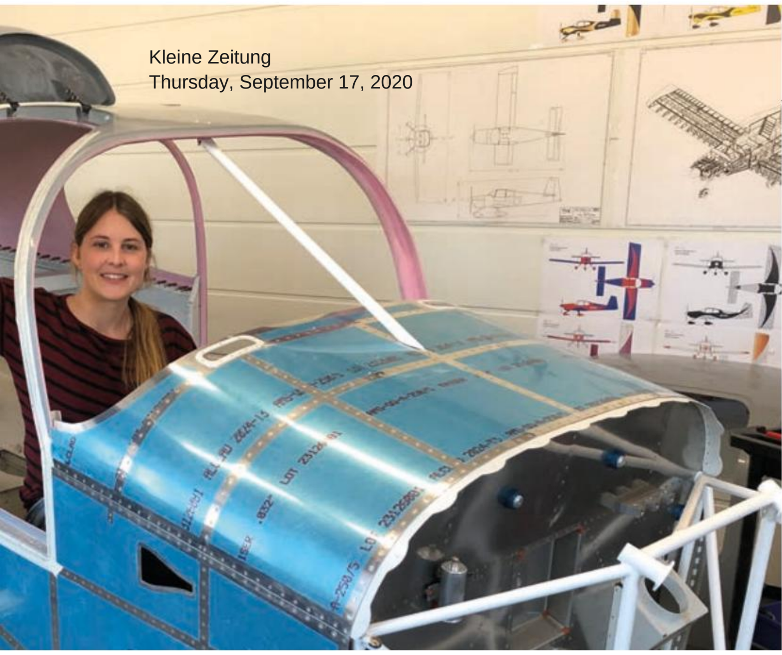

Kleine Zeitung
Thursday, September 17, 2020

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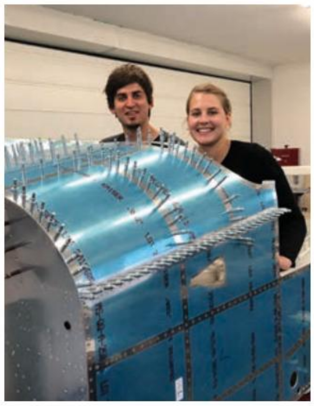
Using a kit plan, the individual parts meticulously precisely installed



Proud of their project: the two Lavanttaler in the RV-10

understand them in all areas That would happen automatically in such a project. After all. are through the Austro Con- troll also constantly tests This was carried out. For this purpose, they were assigned a construction supervisor from the Etrich Clubs, an amateur-

aircraft construction club in Austria, placed alongside. First through a final acceptance of the Austro Control (planned for 2021/ 2022) and after completion a 50-hour trial program to test all parameters and the attached

final flight test through Austro Control, is the aircraft was approved and Ready for deployment. At the airfield in St. Marein - according to the plan - should The machine will be stationed there. Parallel to the construction project Do the two have a pilot?

training to the extent of 45 Completed flying lessons in Graz. With the machine they can then at least with one tank to cover a range of 1328 kilometers – 1328 kilometers towards freedom, the Dreaming felt like a tailwind.

ADVERTISEMENT

A request to our authors

We received this newspaper article, and while compiling the IE Impulse, we suddenly discovered that the photos are of terrible quality. Presumably, the article was copied from the Kleine Zeitung website.

This is often read on a smartphone, and the few pixels are sufficient for that. Therefore, we asked Christian for a few more photos in full resolution; after all, they show all the details that interest us, and you can also recognize the people.

We had a similar experience with several pictures from our summer get-together in Krems: all nice, but rather blurry. They must have been processed by WhatsApp before reaching us. WhatsApp downsizes images; otherwise, it would be impossible to send so many at once, as is currently happenin[illegible] world wide. Furthermore, image quality isn't a major concern on this and other "social" media platforms, since it's mostly about food, cats, and fisheye-distorted selfies anyway. And unfortunately, computers are not yet able to reliably restore the sharpness of an image after it's been defaced.

Unfortunately, we only received a PowerPoint presentation from the meeting in Switzerland. However, through diligent back-and-forth emailing, we finally managed to get some good recordings. Thanks to everyone who helped.

And we have a request: Anyone wishing to contribute to IE Impulse (which is always welcome) should send the **text as a Word file** and the **images separately as full-size .jpeg files as** attachments. Each image should include a line with the names of the people pictured, and somewhere in the text, please also indicate who took the photos or at least where they were taken.

From September 14th to 20th.
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Compared to the demands we place on photos, text is less critical. It can be clunky, error-ridden, grammatically substandard – all of which can be easily fixed.

It's also perfectly fine to send us successful photos and texts, even if the next newspaper isn't due for a long time. Everything ends up in a folder, and when putting the newspaper together, we're always glad to have a wealth of material to choose from.

Christoph Canaval

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For sale:

FLYBOX PR1-P propeller controller, for Rotax with electric adjustable propeller.
Original packaging, unused, for €450 (new price €650).

The controller is compatible with virtually all electric variable-pitch propellers on Rotax engines, as the device is programmable. An external potentiometer can be connected to adjust the propeller speed.
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Private advertisements are free for club members, for non-members €7 per advertisement.

Commercial advertisements in exchange for a donation.
Permanent ads appear until revoked, others appear depending on the order, 3 times for members.

The **"pilots' regular meeting"**
The pilots' get-together is held every first Saturday of the month at Hofkirchen airfield. If the first Saturday of the month falls on a public holiday, the meeting takes place a week later. We meet at approximately 6 p.m. in the cockpit café for a chat, some petrol-related discussions, and an exchange of experiences.


Ready for the next flying season

All IEC workshops and training courses are also held at Hofkirchen airfield (in the seminar room).

Hans Brandstätter also brings the IEC's **electronic scales** to non-members for a fee to cover expenses.

Contribution towards expenses per aircraft: 50 euros.
johann.brandstaetter17@gmail.com
+43 664 22 77 564

Club jackets made of blue fleece, printed with "Igo Etrich Club Austria", are available in all sizes from Heidi Wolf for €20. They are comfortable to wear, very warm, extremely practical, and above all, lightweight.

They can also be obtained from Heidi or Othmar at the regulars' table in Hofkirchen.

Austrian Post AG – Info.Mail Postage paid
Publishing post office Wieselburg.
Undeliverable copies to: Othmar Wolf, 3252 Petzenkirchen, Dürnbachgasse 2